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Cambridge Assessment International Education

Cambridge International AS & A Level

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COMPUTER SCIENCE **9618/32**

Paper 3 Advanced Theory **October/November 2024**

1 hour 30 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS
- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **16** pages. Any blank pages are indicated.

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1 (a) Describe how packet switching is used to transmit messages across a network.

- In packet switching, a message is divided into smaller packets
- Each packet is given a header containing info. like, Source IP add, destination IP add, & packet NO.
- The packets travel independently through the network. using different paths.
- A destination packets are reordered and reassembled.

[3]

(b) State **two** benefits and **two** drawbacks of packet switching as a method of transmitting messages across a network.

Benefit 1 Packet switching allows packets to be re-routed if there is congestion or failure in the network.

Benefit 2 Network bandwidth is used more efficiently b/c packets from diff. comms can share the same path.

Drawback 1 Packets may arrive out of order and need to be reassembled, causing delay

Drawback 2 Packet switching requires complex routing algorithms and significant RAM at the routers.

[4]

2 (a) Describe **serial file organisation** as a method of storing data records in a file.

In serial file organisation, data records are stored one after the other in the order in which they are collected. New records are simply appended at the end of the file, without any sorting. [2]

(b) State **one** example of a use for serial file organisation.

For storing data logging files, such as financial transactions committed over time. [1]

3 (a) Describe the user-defined data type **record**.

- It is a composite data type that groups together multiple related items of data into a single structure.
- Each item in a record can have a different datatype.
- A record allows the different pieces of info that describe one entity to be stored together.

[3]

(b) A programmer defines a record, `Order`, to store the following data:

- account number
- order number
- order price
- order date.

Write **pseudocode** statements to define this record.

```
TYPE Order
    DECLARE AccountNumber : INTEGER
    DECLARE OrderNumber : INTEGER
    DECLARE OrderPrice : REAL
    DECLARE OrderDate : DATE
ENDTYPE.
```

[4]

4 Numbers are stored in a computer using binary floating-point representation with:

- 12 bits for the mantissa
- 4 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Calculate the denary value of the given normalised binary floating-point number.

Show your working.

$\frac{1}{2^1}$ $\frac{1}{2^2}$ $\frac{1}{2^3}$ $\frac{1}{2^4}$ $\frac{1}{2^5}$ $\frac{1}{2^6}$

Mantissa

-1 .5 .25 .125 .0625

0	1	0	0	0	1	1	1	0	1	1	1

Exponent

-8 4 2 1

0	1	1	1

Working 0.10001110111 $\times 2^7$ .5 .25 .125 .0625

1000111.0111 .0 1 1 1

+71 .4375

1 2 4 64 71

.25 .125 .0625 .4375

Answer ....

[2]

(b) Calculate the normalised binary floating-point representation of –49.1875 in this system.

Show your working.

Mantissa

1	0	0	1	1	1	0	1	1	0	1	0

Exponent

0	1	1	0

Working –49.1875

+49.1875

110001.0011 .125 .0625

0.1100010011 $\times 2^6$ .1875

01100010011 0110 +ve

10011101101 0110 -ve

[4]

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5 (a) Name and describe **two** protocols used by the Application Layer of the TCP/IP protocol suite.

Protocol 1 HTTP (Hypertext Transfer Protocol)

Description HTTP is used to transfer web pages and other hypermedia resources b/w the webserver and a client (browser). It defines how msgs are formatted and transmitted.

Protocol 2 SMTP (Simple Mail Transfer Protocol)

Description SMTP is used to send emails from an email client to an email server, and also to transfer emails b/w servers.

[4]

(b) Explain the purpose and function of the Application Layer in the TCP/IP protocol suite.

- AL provides an interface b/w user app and the underlying network services.
- It defines protocols that allow apps to request specific network functions, such as sending emails or tranfering files.
- It prepares data for transmission to the lower layers and also interprets incoming data for the user or app.

[3]

6 The truth table for a logic circuit is shown.

Minterm

INPUT				OUTPUT
A	**B**	**C**	**D**	**X**
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

$\bar{A}\bar{B}\bar{C}D$
$\bar{A}\bar{B}C\bar{D}$
$\bar{A}B\bar{C}D$
$\bar{A}BC\bar{D}$
$A\bar{B}\bar{C}D$
$A\bar{B}C\bar{D}$
$AB\bar{C}D$
$ABC\bar{D}$

(a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

X = $\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D +$
$A\bar{B}C\bar{D} + AB\bar{C}D + ABC\bar{D}$ [3]

(b) Complete the Karnaugh map (K-map) for the given truth table.

CD \ AB	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

[2]

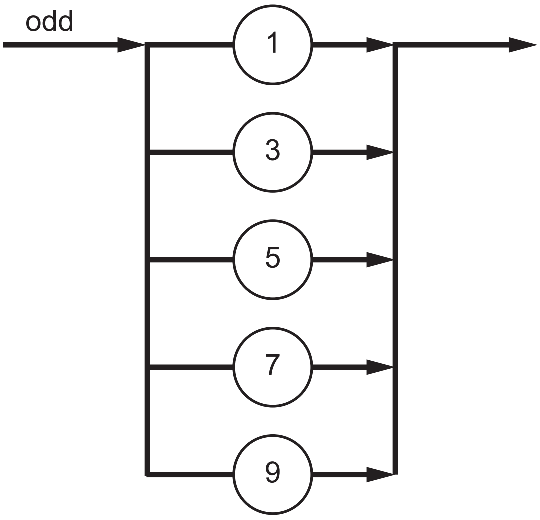
(c) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

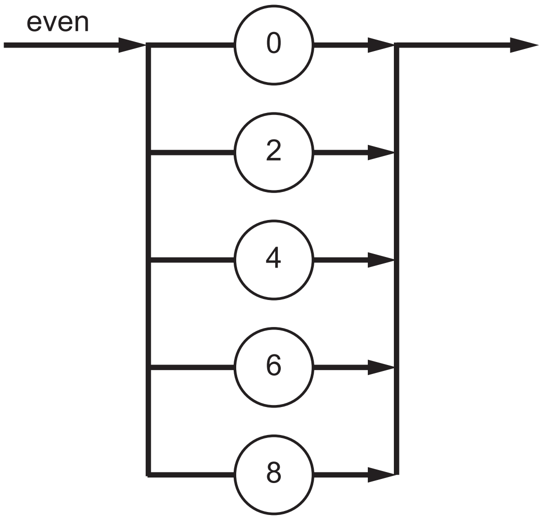
(d) Write the Boolean logic expression from your answer to part **(c)** as the simplified sum-of-products.

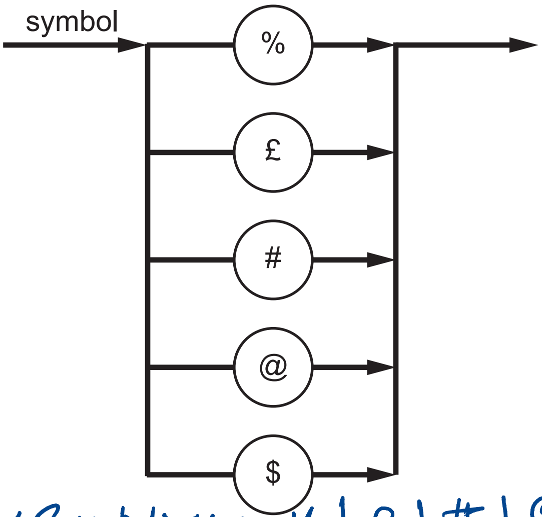
X = $\bar{C}D + C\bar{D}$ ....

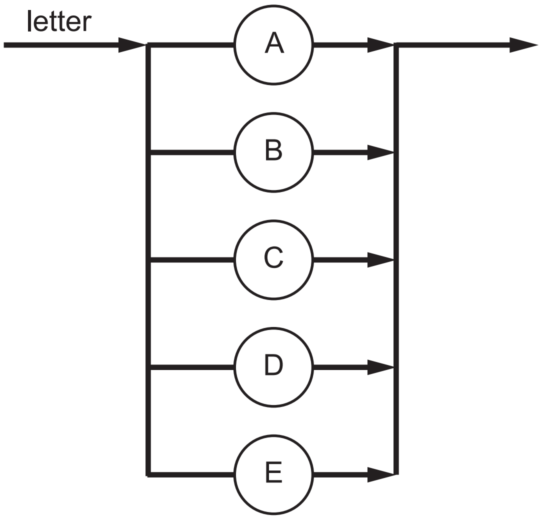
.... [2]

7 Several syntax diagrams are shown.

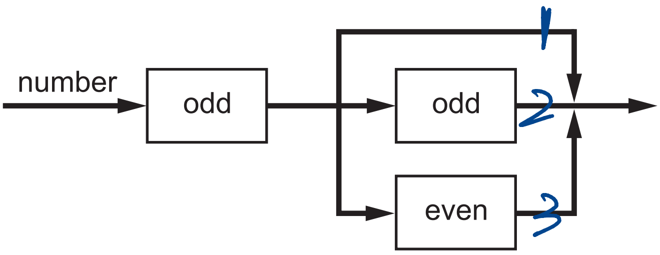








<Symbol> ::= % | £ | # | @ | $



<number> ::= <odd> | <odd><odd> | <odd><even>

(a) State why each number is invalid for the given syntax diagrams.

`21`

Reason ....

....

`123`

Reason ....

.... [2]

(b) Complete the Backus-Naur Form (BNF) for the given syntax diagrams.

`<symbol> ::=` ....

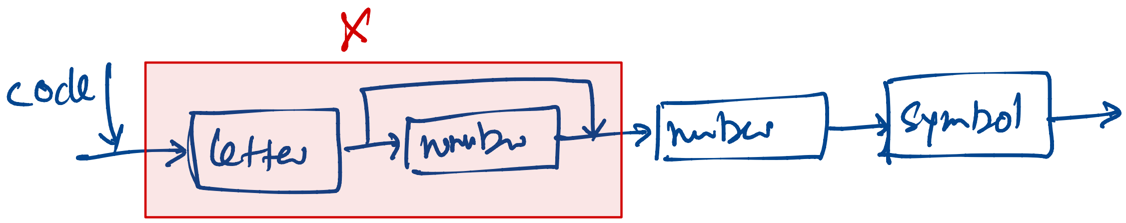
....

`<number> ::=` ....

.... [2]

(c) A new syntax rule, **code**, is required. It must begin with a letter, followed by one or two numbers, and end with a symbol.

(i) Draw a syntax diagram for **code**.



[3]

(ii) Write the BNF for **code**.

<Code> ::= <X> <number><symbol>

<X> ::= <letter> | <letter><number>

....

.... [2]

8 Complex Instruction Set Computer (CISC) is a type of processor.

Identify **four** features of a CISC processor.

1 Have a large & complex inst. set.

2 A single CISC inst. can perform multiple ops.

3 Have variable lengths.

4 Have an complex inst. decoder built into the H/W.

[4]

9 (a) The kernel is the central component of an Operating System (OS).

Outline how the kernel of an OS acts as an interrupt handler.

- Interrupt received
- Current process is paused and saves its states.
- Kernel runs an interrupt service routine to handle the interrupt
- Restores prev. task & resumes.

[2]

(b) (i) State what is meant by the term **multi-tasking** in an Operating System.

Ability of an O/S to run multiple programs apparently at the same time.

[1]

(ii) Describe how multi-tasking is implemented in an Operating System.

[2]

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10 Objects and classes form the basic structure of Object-Oriented Programming (OOP).

(a) Outline the structure of a class.

- Attributes
- Methods
- Access modifiers
- Constructor.

[3]

(b) Give **three** differences between an object and a class.

1 Class | object

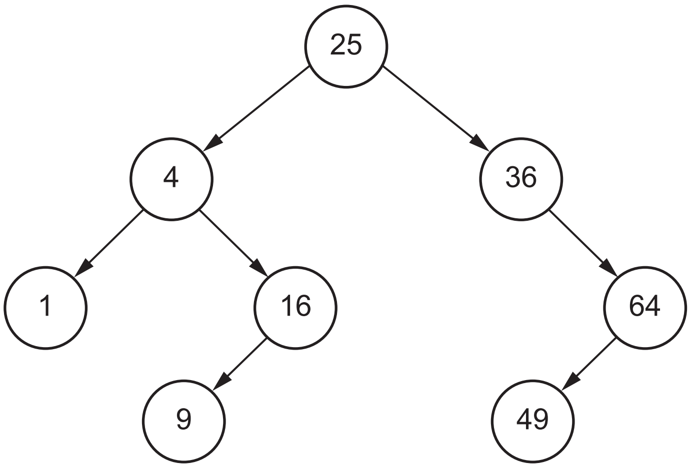
Blueprint | instance

2 doesn't occupy RAM | occupies RAM

3 Defines structure & behaviour | Stores actual values and executes methods.

[3]

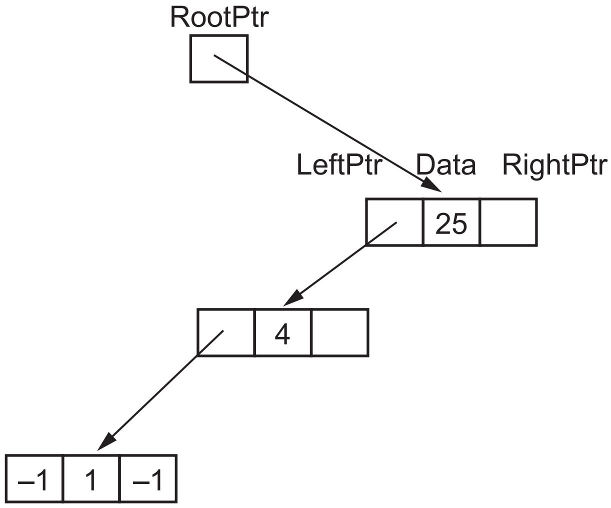
11 This binary tree shows an ordered list of integers.



(a) A linked list of nodes is used to store the data. Each node consists of a left pointer, the data and a right pointer.

−1 is used to represent a null pointer.

Complete this linked list to represent the given binary tree organisation.



[4]

(b) A 2D array is used to store the nodes of the linked list in part **(a)**.

Complete the diagram using your answer for part **(a)**.

RootPtr

0

FreePtr

Index	LeftPtr	Data	RightPtr
0		25	
1		4	
2		36	
3		1	
4		16	
5		64	
6		9	
7		49	
8			

[4]

(c) The linked list in part **(a)** is implemented using a 1D array of records. Each record contains a left pointer, data and a right pointer.

The following pseudocode represents a function that searches for an element in the array of records `LinkList`. It returns the index of the record if the element is found, or it returns a null pointer if the element is not found.

Complete the pseudocode for the function.

```
FUNCTION SearchList(Item : INTEGER) ..........................................
   NullPtr ← -1

   .................................................. ← RootPtr
   WHILE NowPtr <> NullPtr
      IF LinkList[NowPtr].Data < Item THEN
         NowPtr ← LinkList[NowPtr].RightPtr
      ELSE

         IF ............................................................ THEN

            NowPtr ← ...................................................
         ELSE
            RETURN NowPtr
         ENDIF
      ENDIF
   ENDWHILE
   RETURN NullPtr
ENDFUNCTION
```

[4]

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